Positive Homogeneous Forms with Nonsingular Hessian

A full affirmative answer to the question in arXiv:1804.00204

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Status. `candidate_full_solution_likely_valid`.

Source question. Friedland–Gaubert ask in Remark 1 after their Theorem 9 (PDF page 12) whether positivity away from the origin together with invertibility of the Hessian away from the origin forces strict convexity for an even-degree homogeneous form [1].

> *Remark* 1. The arguments of the proof of the Theorem 9 apply if we replace the condition *3* of Theorem 9 by the condition that the Hessian of $F(\mathbf{x})$ is invertible for each $\mathbf{x} \neq \mathbf{0}$. For $d = 2$, the condition *2*, i.e., (5.3), yields that F is strictly convex. We do not know if the condition (5.3) and the condition that the Hessian of $F(\mathbf{x})$ is invertible implies that F is strictly convex for an even $d > 2$.

Answer. Yes. In fact the Hessian must be positive definite at every nonzero point, and none of the surrounding tensor nonnegativity or irreducibility assumptions is needed.

Proof intuition. Minimize the form on the unit sphere. At a minimizer, the constrained second-derivative test makes the Hessian positive in every tangent direction. Homogeneity supplies a positive radial eigenvalue and kills the radial–tangential cross terms, so the Hessian is positive definite at this one point. A continuous nonsingular symmetric matrix cannot change its inertia along a path. Since punctured Euclidean space is connected in dimension at least two, positivity propagates everywhere.

Theorem 1. *Let $F : \mathbb{R}^n \to \mathbb{R}$ be a homogeneous polynomial of degree $d > 1$. Assume*

$$F(x) > 0 \quad (x \neq 0), \qquad \det \nabla^2 F(x) \neq 0 \quad (x \neq 0).$$

Then d is even, $\nabla^2 F(x)$ is positive definite for every $x \neq 0$, and F is strictly convex on $\mathbb{R}^n$.

Proof. Positivity at both x and $-x$ forces the polynomial degree d to be even. First suppose $n \geq 2$. Compactness of the unit sphere gives a point $u \in S^{n-1}$ at which $F|_{S^{n-1}}$ is minimal. Put $m = F(u) > 0$. The Lagrange-multiplier equation and Euler's identity give

$$\nabla F(u) = \lambda u, \qquad \lambda = u^T \nabla F(u) = dF(u) = dm > 0. \tag{1}$$

The constrained second-derivative test says that, for every $v \perp u$,

$$v^T \big(\nabla^2 F(u) - \lambda I\big) v \geq 0, \qquad \text{hence} \qquad v^T \nabla^2 F(u) v \geq \lambda \|v\|^2. \tag{2}$$

Differentiating Euler's identity $x^T \nabla F(x) = dF(x)$ yields

$$\nabla^2 F(x) x = (d-1) \nabla F(x). \tag{3}$$

At u, equations (1)–(3) show that u is a Hessian eigenvector with eigenvalue $(d-1)\lambda > 0$. They also show that $u^T \nabla^2 F(u) v = 0$ whenever $v \perp u$. Thus, for $a \in \mathbb{R}$ and $v \perp u$,

$$(au + v)^T \nabla^2 F(u)(au + v) \geq (d-1)\lambda a^2 + \lambda \|v\|^2,$$

which is positive unless $a = 0$ and $v = 0$. Therefore $\nabla^2 F(u)$ is positive definite.

The inertia of a continuous family of real symmetric nonsingular matrices is locally constant: an eigenvalue would have to pass through zero in order to change sign. The set $\mathbb{R}^n \setminus \{0\}$ is connected for $n \geq 2$, so the inertia of $\nabla^2 F(x)$ is constant there. Since it is $(n, 0, 0)$ at u, $\nabla^2 F(x)$ is positive definite at every nonzero x.

If $n = 1$, homogeneity gives $F(x) = ax^d$ with $a > 0$ and even d, so $F''(x) = ad(d-1)x^{d-2} > 0$ for $x \neq 0$; the same conclusion holds.

Finally take distinct $x, y \in \mathbb{R}^n$ and set $g(t) = F((1-t)x + ty)$ for $0 \leq t \leq 1$. Then

$$g''(t) = (y - x)^T \nabla^2 F((1-t)x + ty)(y - x) > 0$$

except possibly at the single value of t for which $(1-t)x + ty = 0$. Consequently g' is strictly increasing, so g is strictly convex. This on every line segment is exactly strict convexity of F. $\square$

Remark 1 (strength and scope)**.** *The proof only needs a C^2 positive homogeneous function on $\mathbb{R}^n \setminus \{0\}$ with a continuous Hessian and the usual behavior at the origin needed for the last line-segment argument. For the polynomial question as printed, all hypotheses are automatic. In particular, tensor nonnegativity, weak irreducibility, and the restriction $d > 2$ play no role.*

Novelty and verification. The key point is the positive-definite Hessian anchor supplied by a sphere minimum, followed by invariance of inertia. Bounded exact-phrase and arXiv searches through 2026-08-13 found the source question and standard results assuming Hessian positivity, but no matching published answer. Novelty confidence is moderate pending specialist review. Every algebraic identity, the constrained second-order test, the connectivity step, and the one-dimensional edge case were checked separately in the accompanying verification report.

Human-review recommendation. Check equations (2) and (3), then verify that nonsingularity makes Hessian inertia constant on $\mathbb{R}^n \setminus \{0\}$. These three lines carry the entire argument.

References

[1] S. Friedland and S. Gaubert, *Spectral inequalities for nonnegative tensors and their tropical analogues*, Vietnam J. Math. 48 (2020), 893–928; arXiv:1804.00204.